1 Introduction

Notation:

- Random variable X_i with value x_i and possible values from $\mathcal{X}$,
- Random vector $\mathbf{X}$ with range $\mathcal{X}$ and some assignment $\mathbf{x}$,
- Set $\mathcal{A}$, and a measure $\mathbb{M}$.

2 Central Distribution

Definition 2.1 (Probabilistic Model)**.** We define a *probabilistic model* $\mathcal{M}$ as a pair $(\mathcal{S}, \mathcal{P})$, where $\mathcal{S}$ is a sample space, and $\mathcal{P}$ is a set of probability measures on $\mathcal{S}$.

Definition 2.2 (Divergence Measure)**.** Let $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \dots\}$ be a set of pairs of discrete random variables in a probabilistic model $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$. We define the *divergence measure of* $P \in \mathcal{P}$ *for* $\mathcal{X}$ as follows:

$$\mathbb{D}_{\mathcal{X}}\left[P\right] = \max_{Q \in \mathcal{P}} \mathbb{E}_Q \left[\sum_i \mathbb{D}\left(Q_{X_i|e_i} \parallel P_{X_i|e_i}\right) \right], \tag{1}$$

where $\mathbb{E}_Q\left[\,\cdot\,\right]$ denotes expectation relative to Q and each $\mathbb{D}\left(Q_{X_i|e_i} \parallel P_{X_i|e_i}\right)$ is the Kullback-Leibler divergence between $P(X_i|e_i)$ and $Q(X_i|e_i)$:

$$\mathbb{D}\left(Q_{X_i|e_i} \parallel P_{X_i|e_i}\right) = \sum_{x_i} Q(x_i \mid e_i) \ln \frac{Q(x_i \mid e_i)}{P(x_i \mid e_i)}\cdot$$

We also define *the central distribution for* $\mathcal{X}$ to be a distribution $C_{\mathcal{X}} \in \mathcal{P}$ which minimizes the divergence measure for $\mathcal{X}$:

$$C_{\mathcal{X}} = \arg\min_{P \in \mathcal{P}} \mathbb{D}_{\mathcal{X}}\left[P\right]\cdot \tag{2}$$

Proposition. Using the linearity of expectation, we have

$$\mathbb{D}_{\mathcal{X}}\left[P\right] = \max_{Q \in \mathcal{P}} \sum_i \sum_{x_i, e_i} Q(x_i, e_i) \ln \frac{Q(x_i \mid e_i)}{P(x_i \mid e_i)}\cdot$$

Definition 2.3 (Parametric Model)**.** Let Θ be a random vector. A parametric model is a probabilistic model $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$, where for every $P \in \mathcal{P}$ there is a prior probability distribution p, such that for every $s \in \mathcal{S}$ we have

$$P(s) = \int \pi(s \mid \theta) p(\theta) d\theta, \tag{3}$$

while π is a known probability distribution.

Lemma 1. *Let $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ be a parametric model, and $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \dots\}$ be a set of discrete random variable pairs in $\mathcal{M}_\Theta$. For every $P \in \mathcal{P}$ we have*

$$\begin{aligned}\mathbb{D}_{\mathcal{X}}[P] &\leq \max_{\theta\in\Theta} \mathbb{E}_\pi \left[\sum_i \mathbb{D}\left(\pi_{X_i|e_i,\theta} \parallel P_{X_i|e_i}\right) \,\middle|\, \theta \right] \\ &= \max_{\theta\in\Theta} \sum_i \sum_{x_i,e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{P(x_i \mid e_i)} \cdot\end{aligned}$$

Proof. Let R be an arbitrary distribution in $\mathcal{M}_\Theta$, since the Kullback-Leibler divergence is positive, its expected value is also positive, and for every parameter value θ we have

$$\sum_i \sum_{x_i,e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{R(x_i \mid e_i)} \geq 0 \cdot$$

Using this, for every distribution P and every parameter θ, we obtain

$$\sum_i \sum_{x_i,e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)} \leq \sum_i \sum_{x_i,e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{P(x_i \mid e_i)} \cdot$$

Since this inequality holds for any θ, the maximum of the right-hand side is also greater than or equal to the maximum of the left-hand side w.r.t. θ:

$$\max_{\theta\in\Theta} \sum_i \sum_{x_i,e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)} \leq \max_{\theta\in\Theta} \mathbb{E}_\pi \left[\sum_i \mathbb{D}\left(\pi_{X_i|e_i,\theta} \parallel P_{X_i|e_i}\right) \,\middle|\, \theta \right] \cdot \tag{4}$$

$\mathcal{M}_\Theta$ is a parametric model and $R(x_i, e_i) = \int \pi(x_i, e_i \mid \theta) r(\theta) d\theta$, where $r(\theta)$ is a probability distribution. Clearly, for every θ we have

$$\sum_i \sum_{x_i,e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)} \leq \max_{\theta\in\Theta} \sum_i \sum_{x_i,e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)} \cdot$$

Since $r(\theta)$ is a distribution and is non-negative, we can multiply this inequality by $r(\theta)$ and integrate. Notice that the right hand side is not a function of θ and it will remain unchanged because $\int r(\theta) d\theta = 1$:

$$\mathbb{E}_R \left[\sum_i \mathbb{D}\left(R_{X_i|e_i} \parallel P_{X_i|e_i}\right) \right] \leq \max_{\theta\in\Theta} \sum_i \sum_{x_i,e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)} \cdot$$

Now from Inequality 4, for *any* distribution P and R in $\mathcal{M}_\Theta$, it follows that

$$\mathbb{E}_R \left[\sum_i \mathbb{D}\left(R_{X_i|e_i} \parallel P_{X_i|e_i}\right) \right] \leq \max_{\theta\in\Theta} \mathbb{E}_\pi \left[\sum_i \mathbb{D}\left(\pi_{X_i|e_i,\theta} \parallel P_{X_i|e_i}\right) \,\middle|\, \theta \right],$$

which proves the theorem. □

Theorem 2. *Let $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ be a parametric model and $\mathcal{X}$ be a set of pairs of discrete random variables in $\mathcal{M}_\Theta$. If for every value of θ there is a distribution $R \in \mathcal{P}$ such that $R(s) = \pi(s \mid \theta)$ for every $s \in \mathcal{S}$, then for any $P \in \mathcal{P}$ we have:*

$$\mathbb{D}_{\mathcal{X}}[P] = \max_{\theta\in\Theta} \mathbb{E}_\pi \left[\sum_i \mathbb{D}\left(\pi_{X_i|e_i,\theta} \parallel P_{X_i|e_i}\right) \,\middle|\, \theta \right]. \tag{5}$$

Proof. For an arbitrary distribution $P \in \mathcal{P}$, we let

$$\theta^* = \arg\max_{\theta\in\Theta} \mathbb{E}_\pi \left[\sum_i \mathbb{D}\left(\pi_{X_i|e_i,\theta} \parallel P_{X_i|e_i}\right) \,\middle|\, \theta \right].$$

In $\mathcal{M}_\Theta$ there should be a distribution $R \in \mathcal{P}$ that equals $\pi(\cdot\,|\theta^*)$, and we can write

$$\mathbb{E}_R \left[\sum_i \mathbb{D}\left(R_{X_i|e_i} \parallel P_{X_i|e_i}\right) \right] = \mathbb{E}_\pi \left[\sum_i \mathbb{D}\left(\pi_{X_i|e_i,\theta^*} \parallel P_{X_i|e_i}\right) \,\middle|\, \theta^* \right].$$

On the other hand, we see

$$\max_{Q\in\mathcal{P}} \mathbb{E}_Q \left[\sum_i \mathbb{D}\left(Q_{X_i|e_i} \parallel P_{X_i|e_i}\right) \right] \geq \mathbb{E}_R \left[\sum_i \mathbb{D}\left(R_{X_i|e_i} \parallel P_{X_i|e_i}\right) \right].$$

Thus,

$$\mathbb{D}_{\mathcal{X}}[P] \geq \max_{\theta\in\Theta} \mathbb{E}_\pi \left[\sum_i \mathbb{D}\left(\pi_{X_i|e_i,\theta} \parallel P_{X_i|e_i}\right) \,\middle|\, \theta \right].$$

This inequality and Lemma 1 prove the theorem. □

Theorem 3 (Minimax theorem [1])**.** *Let $\mathcal{X} \subset \mathbb{R}^n$ and $\mathcal{Y} \subset \mathbb{R}^m$ be compact convex sets. If $f : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$ is a continuous function that is concave-convex, i.e.*

- *$f(\cdot, y) : \mathcal{X} \to \mathbb{R}$ is concave for fixed y, and*
- *$f(x, \cdot) : \mathcal{Y} \to \mathbb{R}$ is convex for fixed x.*

Then we have that

$$\max_{x\in\mathcal{X}} \min_{y\in\mathcal{Y}} f(x, y) = \min_{y\in\mathcal{Y}} \max_{x\in\mathcal{X}} f(x, y). \tag{6}$$

Theorem 4. *Consider a generic parametric model $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$, and let $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \dots\}$ be a set of pairs of discrete random variables in $\mathcal{M}_\Theta$. Assume that for every $Q_1, Q_2 \in \mathcal{P}$ and any real number $0 \leq \alpha \leq 1$, there is a distribution $Q^* \in \mathcal{P}$, such that for every i, x_i, e_i we have*

$$Q^*(x_i \mid e_i) = \alpha Q_1(x_i \mid e_i) + (1-\alpha) Q_2(x_i \mid e_i)\cdot \tag{7}$$

In this model, the central distribution for $\mathcal{X}$ can be obtained by:

$$\begin{aligned} C_{\mathcal{X}} &= \arg\max_{P\in\mathcal{P}} \int \mathbb{E}_\pi \left[\sum_i \mathbb{D}\left(\pi_{X_i|e_i,\theta} \parallel P_{X_i|e_i}\right) \,\middle|\, \theta \right] p(\theta) d\theta \\ &= \arg\max_{P\in\mathcal{P}} \int \sum_i \sum_{x_i, e_i} P(x_i, e_i, \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{P(x_i \mid e_i)} d\theta, \end{aligned}$$

where $P(x_i, e_i, \theta) = \pi(x_i, e_i \mid \theta) p(\theta)$.

Proof. We prove this theorem using the Minimax theorem. For that, we define a functional as follows:

$$f(r, P_{X_i|e_i}) = \int \mathbb{E}_\pi \left[\sum_i \mathbb{D}\left(\pi_{X_i|e_i,\theta} \parallel P_{X_i|e_i}\right) \,\middle|\, \theta \right] r(\theta) d\theta,$$

where $r(\theta)$ is an arbitrary probability distribution, and $P \in \mathcal{P}$. Since $r(\theta)$ can be any distribution, maximizing with respect to r is like maximizing with respect to θ, and we have

$$C_{\mathcal{X}} = \arg\min_{P\in\mathcal{P}} \max_r f(r, P_{X_i|e_i})\cdot$$

The domain of r is the set of all probability distributions, which is a compact convex set. Moreover, from Equation 7, it follows that the set of probability distributions for $X_i|E_i$ in $\mathcal{M}_\Theta$ is also a compact convex set. Thus, the domain of $f(r, P_{X_i|e_i})$ is a compact convex set.

For a fixed $P_{X_i|e_i}$, $f(\cdot, P_{X_i|e_i})$ is a linear function, which is both concave and convex. On the other hand, if we let $R(x_i, e_i) = \int \pi(x_i, e_i \mid \theta) r(\theta) d\theta$ for $f(r, \cdot)$ we have

$$f(r, \cdot) = c - \sum_i \sum_{x_i, e_i} R(x_i, e_i) \ln P(x_i \mid e_i)\cdot \tag{8}$$

If we calculate the Hessian matrix of $f(r, \cdot)$, we observe that the matrix is diagonal, and each element of the main diagonal is $\frac{R(x_i|e_i)}{P(x_i|e_i)^2}$. Thus, the hessian matrix of $f(\cdot, P_{X_i|e_i})$ is positive semi-definite and $f(r, \cdot)$ is a convex function. Now, Theorem 3 implies

$$\min_{P\in\mathcal{P}} \max_r f(r, P_{X_i|e_i}) = \max_r \min_{P\in\mathcal{P}} f(r, P_{X_i|e_i})\cdot \tag{9}$$

From Equation 8 we can see that for an arbitrary fixed r we have

$$\min_{P\in\mathcal{P}} f(r, P_{X_i|e_i}) = \max_{P\in\mathcal{P}} \sum_i \sum_{x_i,e_i} R(x_i, e_i) \ln P(x_i \mid e_i)\cdot$$

We know that the Kullback–Leibler divergence, and therefore its expected value, is non-negative:

$$\mathbb{E}_R \left[\sum_i \mathbb{D}\left(R_{X_i|e_i} \parallel P_{X_i|e_i} \right) \right] \geq 0\cdot$$

By simple algebra we get:

$$\sum_i \sum_{x_i,e_i} R(x_i, e_i) \ln P(x_i \mid e_i) \leq \sum_i \sum_{x_i,e_i} R(x_i, e_i) \ln R(x_i \mid e_i)\cdot$$

Recall that $\mathcal{M}_\Theta$ is a generic parametric model, so $R \in \mathcal{P}$, and we can reach this upper bound by letting $P = R$.

$$\min_{P\in\mathcal{P}} f(r, P_{X_i|e_i}) = \int \sum_i \sum_{x_i,e_i} R(x_i, e_i, \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{R(x_i \mid e_i)} d\theta\cdot$$

By Equation 9 we conclude:

$$C_{\mathcal{X}} = \arg\max_{R\in\mathcal{R}} \int \mathbb{E}_\pi \left[\sum_i \mathbb{D}\left(\pi_{X_i|e_i,\theta} \parallel R_{X_i|e_i} \right) \middle| \theta \right] r(\theta) d\theta\cdot$$

□

References

[1] Ding-Zhu Du. *Minimax and Its Applications*, pages 339–367. Springer US, Boston, MA, 1995.